job_13-Evaluation report

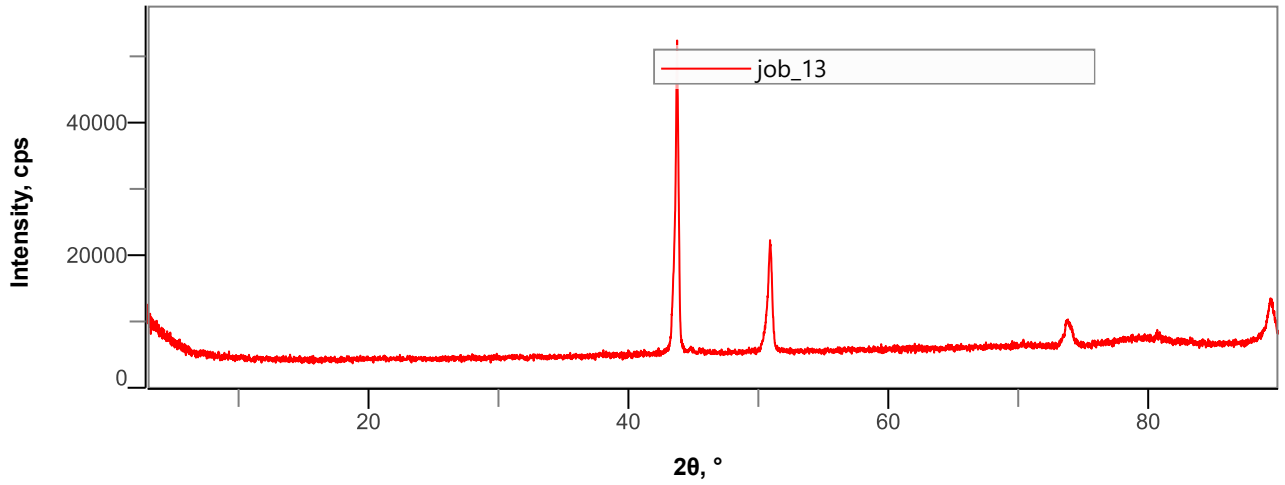
General information

Analysis date	2025-12-17 10:45:55	Measurement start time	2025-11-23 11:40:30
Analyst	Administrator	Operator	Administrator
Sample name	job_13	Comment	
Measured data name	C:\Users\DELL\Desktop\251\Vasanth\job_13.rasx	Memo	

Measurement Conditions

X-Ray generator	40 kV, 15 mA	Scan mode	1D(scan)
Incident primary	No unit	Scan speed/Duration time	10.00 °/min
Goniometer	MiniFlex 300/600	Step width	0.01 °
Attachment	ASC-8	Scan axis	θ/2θ
Filter	None	Scan range	3 ~ 90 °
Selection slit	None	DS	1.25deg
Diffracted beam mono	None	IHS	10 mm
Detector	HyPix-400 MF	SS	Open
Optics attribute	None	RS	Open

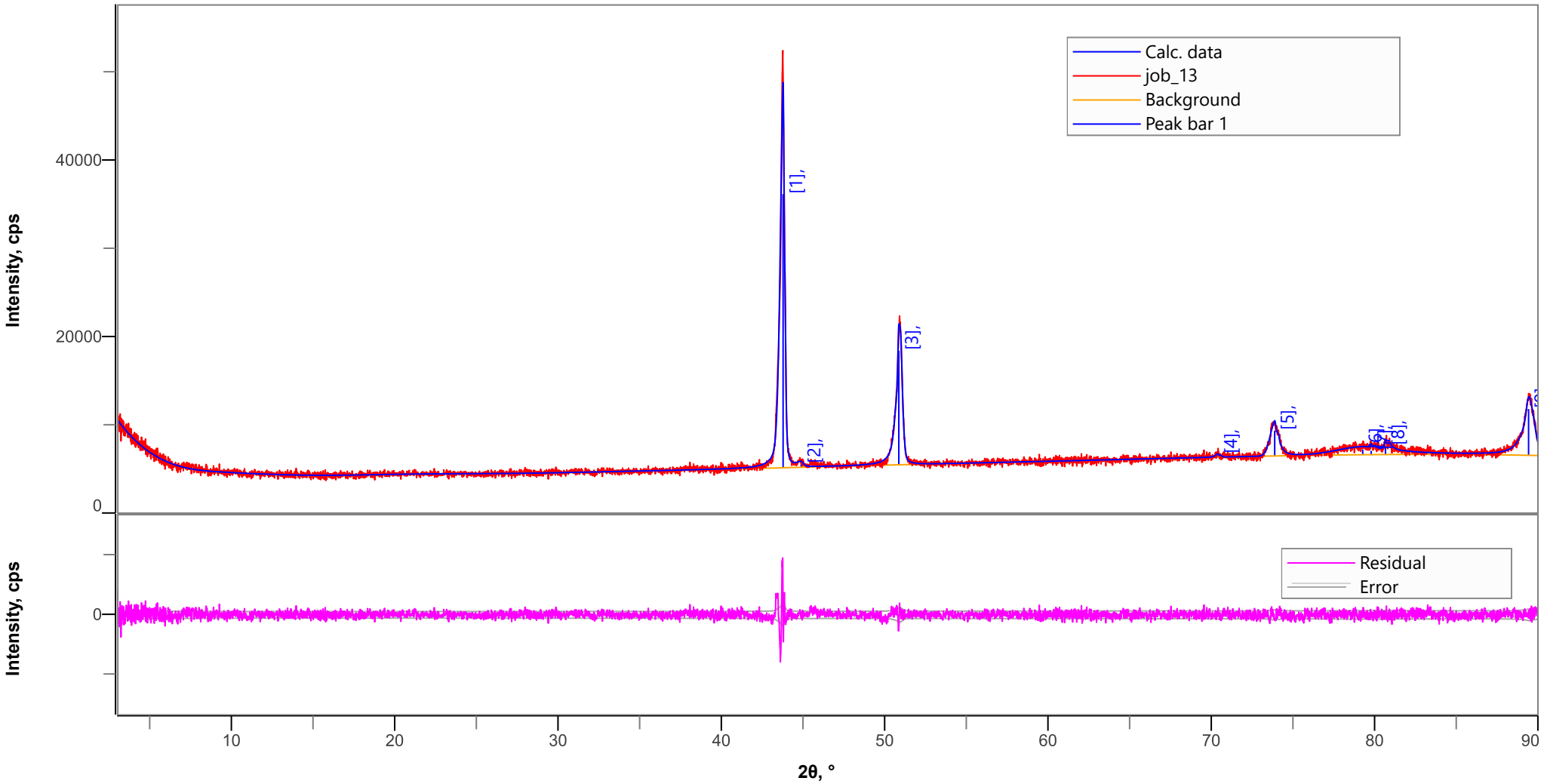
Measured profile view



Peak evaluation conditions

Peak search method	Second derivative method	σ cut	3.00		
Profile fitting	Run completed	Peak shape	Split pseudo-Voigt	Fitting condition	Auto(Refine background)

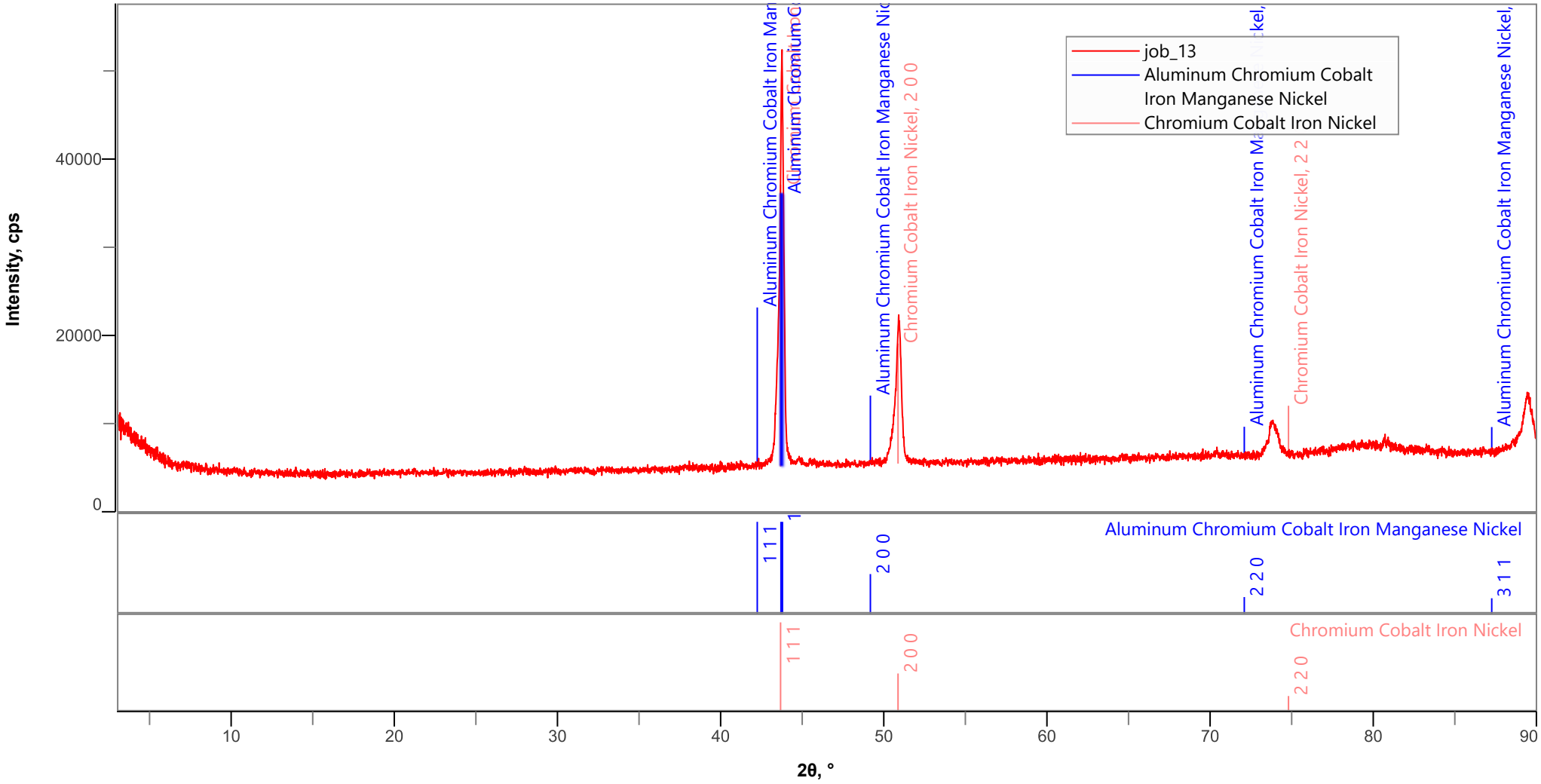
Peak Profile View



Qualitative Analysis Results

Phase name	Chemical formula	FOM	Phase reg. detail	Space Group	DB Card Number
Aluminum Chromium Cobalt Iron Manganese Nickel	Cr0.18 Mn0.18 Fe0.18 Co0.19 Ni0.18 Al0.09	3.298	S/M:PDF-5+ 2026	225 : Fm-3m	04-028-0214

Phase Data View



Peak list

No.	2θ, °	d, Å	Height, cps	FWHM, °	Int. I., cps°	Int. W., °	Asymmetry	Decay (ηL/mL)	Decay (ηH/mH)	Size, Å	Phase Name	Chemical Formula	Card No	Norm. I.	Profile Type	Distribution R.S.D.	Strain, %	Degree of Orientation	Ring Factor	β Cluster	Custom Label
1	43.760 (2)	2.066698 (9)	30946 (555)	0.286 (2)	11246(44)	0.363(8)	3.53 (16)	0.414 (13)	0.78(5)	313(2)	Aluminum Chromium Cobalt Iron Manganese Nickel: 1 1 1	Cr0.18 Mn0.18 Fe0.18 Co0.19 Ni0.18 Al0.09	04-028-0214	100.00	Split Pseudo-Voigt	-	-	-	-	-	
2	44.87 (2)	2.0185 (10)	386 (21)	0.51 (8)	250 (28)	0.65 (11)	3.53 (16)	0.414 (13)	0.78(5)	177(26)	Unknown			2.22	Split Pseudo-Voigt	-	-	-	-	-	
3	50.842 (3)	1.794443 (10)	12909 (371)	0.272 (5)	5617 (40)	0.435 (16)	0.74(4)	1.55 (4)	0.38(4)	338(6)	Aluminum Chromium Cobalt Iron Manganese Nickel: 2 0 0	Cr0.18 Mn0.18 Fe0.18 Co0.19 Ni0.18 Al0.09	04-028-0214	49.94	Split Pseudo-Voigt	-	-	-	-	-	
4	70.38 (5)	1.3367 (9)	325 (16)	0.25 (12)	116 (31)	0.36 (11)	3(3)	0.6 (11)	1(2)	413 (194)	Unknown			1.03	Split Pseudo-Voigt	-	-	-	-	-	
5	73.845 (10)	1.282225 (14)	2866 (116)	0.488 (13)	1875 (73)	0.65 (5)	2.8 (2)	0.00 (14)	1.55(12)	213(6)	Unknown			16.67	Split Pseudo-Voigt	-	-	-	-	-	
6	79.2 (2)	1.208 (3)	603 (28)	3.14 (18)	2826 (205)	4.7 (6)	1.5 (3)	0.0 (4)	1.5(3)	34.3(19)	Unknown			25.13	Split Pseudo-Voigt	-	-	-	-	-	
7	79.75 (3)	1.2015 (3)	427 (19)	0.16 (9)	136 (72)	0.32 (18)	0.5 (3)	1.1 (6)	1.5(3)	667 (375)	Unknown			1.21	Split Pseudo-Voigt	-	-	-	-	-	
8	80.62 (3)	1.1907 (3)	1011 (44)	0.25 (4)	489 (46)	0.48 (7)	0.5 (3)	1.1 (6)	1.5(3)	440(74)	Unknown			4.34	Split Pseudo-Voigt	-	-	-	-	-	
9	89.282	1.0952	5238	0.52	4468 (53)	0.85 (4)	1.23	1.32 (7)	0.67(11)	223(10)	Unknown			39.72	Split Pseudo	-	-	-	-	-	

Lattice parameters

Phase name	a, Å	b, Å	c, Å	α, °	β, °	γ, °
Aluminum Chromium Cobalt Iron Manganese Nickel	3.58308	3.58308	3.58308	90.000	90.000	90.000

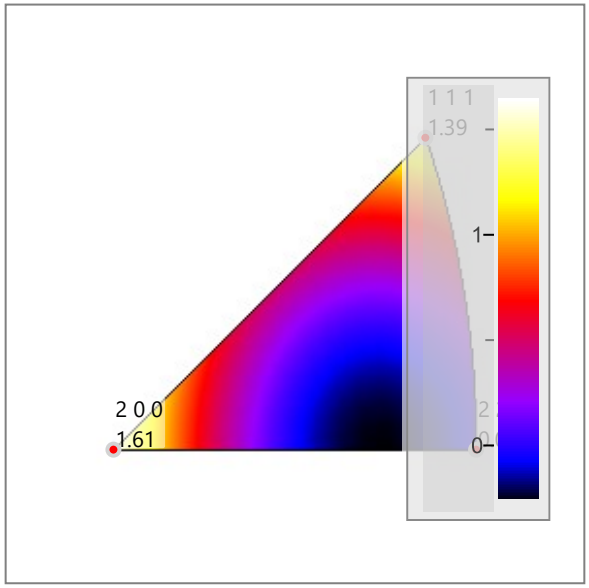
d-I List

Aluminum Chromium Cobalt Iron Manganese Nickel

No.	2θ, °	d, Å	h k l	Norm. I.
1	43.72243	2.06869	1 1 1	100.00
2	50.93053	1.79154	2 0 0	45.08
3	74.89882	1.26681	2 2 0	22.76

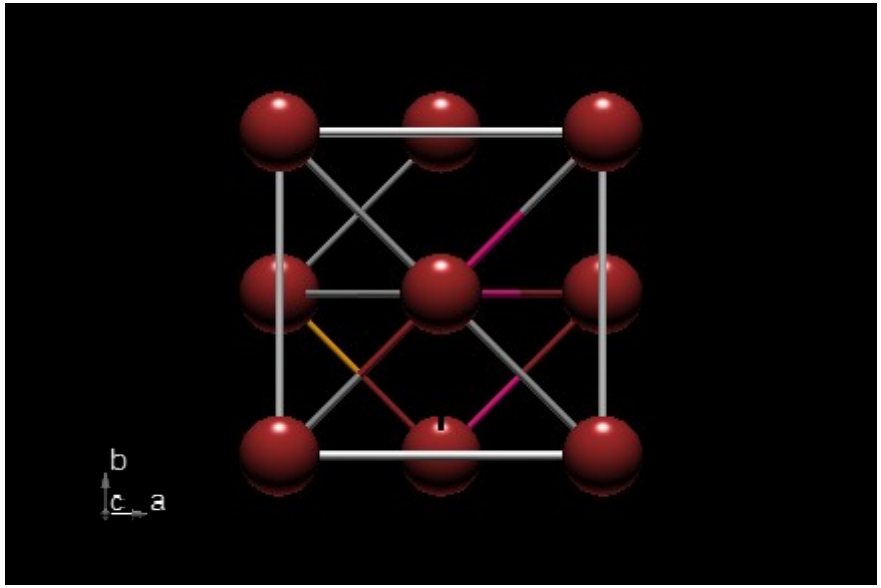
Simple Inverse Pole Figure

Aluminum Chromium Cobalt Iron Manganese Nickel



Crystal Structure

Aluminum Chromium Cobalt Iron Manganese Nickel



Structure parameters

Element	x	y	z	Occ.	B	Charge	Multiplicity	Valence	Bond Valence Sum
Co1(Co)	0.000000	0.000000	0.000000	0.190	0.500	0	4	0.000	0.000
Cr1(Cr)	0.000000	0.000000	0.000000	0.180	0.500	0	4	0.000	0.000
Fe1(Fe)	0.000000	0.000000	0.000000	0.180	0.500	0	4	0.000	0.000
Mn1(Mn)	0.000000	0.000000	0.000000	0.180	0.500	0	4	0.000	0.000
Ni1(Ni)	0.000000	0.000000	0.000000	0.180	0.500	0	4	0.000	0.000
Al1(Al)	0.000000	0.000000	0.000000	0.090	0.500	0	4	0.000	0.000

Distance

Bond Def.	Distance, Å
Ni1-Al1(1/2+x,y,-1/2+z)	2.53362
Ni1-Al1(x,1/2+y,1/2+z)	2.53362
Ni1-Al1(x,1/2+y,-1/2+z)	2.53362
Ni1-Al1(x,-1/2+y,1/2+z)	2.53362
Ni1-Al1(-1/2+x,y,1/2+z)	2.53362
Ni1-Al1(1/2+x,y,1/2+z)	2.53362
Ni1-Al1(1/2+x,-1/2+y,z)	2.53362
Ni1-Al1(-1/2+x,1/2+y,z)	2.53362
Ni1-Al1(1/2+x,1/2+y,z)	2.53362
Co1-Ni1(1/2+x,y,-1/2+z)	2.53362
Co1-Ni1(x,1/2+y,1/2+z)	2.53362
Co1-Ni1(-1/2+x,y,1/2+z)	2.53362
Co1-Ni1(x,1/2+y,-1/2+z)	2.53362
Co1-Ni1(1/2+x,y,1/2+z)	2.53362
Co1-Ni1(1/2+x,-1/2+y,z)	2.53362
Co1-Ni1(-1/2+x,1/2+y,z)	2.53362
Co1-Ni1(1/2+x,1/2+y,z)	2.53362
Co1-Ni1(x,-1/2+y,1/2+z)	2.53362
Co1-Co1(x,-1/2+y,1/2+z)	2.53362
Co1-Co1(x,1/2+y,-1/2+z)	2.53362
Co1-Co1(1/2+x,1/2+y,z)	2.53362
Co1-Co1(-1/2+x,1/2+y,z)	2.53362
Co1-Co1(1/2+x,-1/2+y,z)	2.53362
Co1-Co1(1/2+x,y,1/2+z)	2.53362
Co1-Co1(1/2+x,y,-1/2+z)	2.53362
Co1-Co1(-1/2+x,y,1/2+z)	2.53362

Distance

Bond Def.	Distance, Å
Co1-Co1(x,1/2+y,1/2+z)	2.53362
Co1-Fe1(1/2+x,y,-1/2+z)	2.53362
Co1-Fe1(-1/2+x,y,1/2+z)	2.53362
Co1-Fe1(1/2+x,-1/2+y,z)	2.53362
Co1-Fe1(-1/2+x,1/2+y,z)	2.53362
Co1-Fe1(1/2+x,1/2+y,z)	2.53362
Co1-Fe1(x,1/2+y,-1/2+z)	2.53362
Co1-Fe1(1/2+x,y,1/2+z)	2.53362
Co1-Fe1(x,-1/2+y,1/2+z)	2.53362
Co1-Fe1(x,1/2+y,1/2+z)	2.53362
Co1-Mn1(1/2+x,-1/2+y,z)	2.53362
Co1-Mn1(1/2+x,y,1/2+z)	2.53362
Co1-Mn1(-1/2+x,y,1/2+z)	2.53362
Co1-Mn1(1/2+x,y,-1/2+z)	2.53362
Co1-Mn1(1/2+x,1/2+y,z)	2.53362
Co1-Mn1(x,1/2+y,-1/2+z)	2.53362
Co1-Mn1(-1/2+x,1/2+y,z)	2.53362
Co1-Mn1(x,1/2+y,1/2+z)	2.53362
Co1-Mn1(x,-1/2+y,1/2+z)	2.53362
Co1-Cr1(x,1/2+y,1/2+z)	2.53362
Co1-Cr1(x,-1/2+y,1/2+z)	2.53362
Co1-Cr1(x,1/2+y,-1/2+z)	2.53362
Co1-Cr1(1/2+x,1/2+y,z)	2.53362
Co1-Cr1(-1/2+x,1/2+y,z)	2.53362
Co1-Cr1(1/2+x,-1/2+y,z)	2.53362
Co1-Cr1(1/2+x,y,-1/2+z)	2.53362

Distance

Bond Def.	Distance, Å
Co1-Cr1(-1/2+x,y,1/2+z)	2.53362
Co1-Cr1(1/2+x,y,1/2+z)	2.53362
Co1-Al1(x,-1/2+y,1/2+z)	2.53362
Co1-Al1(x,1/2+y,-1/2+z)	2.53362
Co1-Al1(x,1/2+y,1/2+z)	2.53362
Co1-Al1(-1/2+x,1/2+y,z)	2.53362
Co1-Al1(1/2+x,-1/2+y,z)	2.53362
Co1-Al1(1/2+x,y,1/2+z)	2.53362
Co1-Al1(-1/2+x,y,1/2+z)	2.53362
Co1-Al1(1/2+x,y,-1/2+z)	2.53362
Co1-Al1(1/2+x,1/2+y,z)	2.53362
Fe1-Ni1(-1/2+x,1/2+y,z)	2.53362
Fe1-Ni1(1/2+x,1/2+y,z)	2.53362
Fe1-Ni1(x,1/2+y,-1/2+z)	2.53362
Fe1-Ni1(x,-1/2+y,1/2+z)	2.53362
Fe1-Ni1(x,1/2+y,1/2+z)	2.53362
Fe1-Ni1(1/2+x,y,-1/2+z)	2.53362
Fe1-Ni1(-1/2+x,y,1/2+z)	2.53362
Fe1-Ni1(1/2+x,y,1/2+z)	2.53362
Fe1-Ni1(1/2+x,-1/2+y,z)	2.53362
Fe1-Fe1(-1/2+x,y,1/2+z)	2.53362
Fe1-Fe1(1/2+x,y,-1/2+z)	2.53362
Fe1-Fe1(1/2+x,y,1/2+z)	2.53362
Fe1-Fe1(1/2+x,-1/2+y,z)	2.53362
Fe1-Fe1(-1/2+x,1/2+y,z)	2.53362
Fe1-Fe1(1/2+x,1/2+y,z)	2.53362

Distance

Bond Def.	Distance, Å
Fe1-Fe1(x,1/2+y,-1/2+z)	2.53362
Fe1-Fe1(x,1/2+y,1/2+z)	2.53362
Fe1-Fe1(x,-1/2+y,1/2+z)	2.53362
Fe1-Mn1(1/2+x,y,-1/2+z)	2.53362
Fe1-Mn1(-1/2+x,y,1/2+z)	2.53362
Fe1-Mn1(1/2+x,y,1/2+z)	2.53362
Fe1-Mn1(1/2+x,-1/2+y,z)	2.53362
Fe1-Mn1(1/2+x,1/2+y,z)	2.53362
Fe1-Mn1(x,1/2+y,-1/2+z)	2.53362
Fe1-Mn1(x,-1/2+y,1/2+z)	2.53362
Fe1-Mn1(x,1/2+y,1/2+z)	2.53362
Fe1-Mn1(-1/2+x,1/2+y,z)	2.53362
Fe1-Al1(-1/2+x,1/2+y,z)	2.53362
Fe1-Al1(1/2+x,-1/2+y,z)	2.53362
Fe1-Al1(-1/2+x,y,1/2+z)	2.53362
Fe1-Al1(1/2+x,y,1/2+z)	2.53362
Fe1-Al1(1/2+x,1/2+y,z)	2.53362
Fe1-Al1(1/2+x,y,-1/2+z)	2.53362
Fe1-Al1(x,1/2+y,-1/2+z)	2.53362
Fe1-Al1(x,-1/2+y,1/2+z)	2.53362
Fe1-Al1(x,1/2+y,1/2+z)	2.53362
Mn1-Ni1(1/2+x,y,1/2+z)	2.53362
Mn1-Ni1(1/2+x,y,-1/2+z)	2.53362
Mn1-Ni1(x,1/2+y,1/2+z)	2.53362
Mn1-Ni1(-1/2+x,y,1/2+z)	2.53362
Mn1-Ni1(x,-1/2+y,1/2+z)	2.53362

Distance

Bond Def.	Distance, Å
Mn1-Ni1(x,1/2+y,-1/2+z)	2.53362
Mn1-Ni1(-1/2+x,1/2+y,z)	2.53362
Mn1-Ni1(1/2+x,-1/2+y,z)	2.53362
Mn1-Ni1(1/2+x,1/2+y,z)	2.53362
Mn1-Mn1(x,-1/2+y,1/2+z)	2.53362
Mn1-Mn1(x,1/2+y,1/2+z)	2.53362
Mn1-Mn1(1/2+x,y,1/2+z)	2.53362
Mn1-Mn1(1/2+x,1/2+y,z)	2.53362
Mn1-Mn1(-1/2+x,y,1/2+z)	2.53362
Mn1-Mn1(1/2+x,y,-1/2+z)	2.53362
Mn1-Mn1(-1/2+x,1/2+y,z)	2.53362
Mn1-Mn1(x,1/2+y,-1/2+z)	2.53362
Mn1-Mn1(1/2+x,-1/2+y,z)	2.53362
Mn1-Al1(-1/2+x,1/2+y,z)	2.53362
Mn1-Al1(x,1/2+y,1/2+z)	2.53362
Mn1-Al1(x,-1/2+y,1/2+z)	2.53362
Mn1-Al1(x,1/2+y,-1/2+z)	2.53362
Mn1-Al1(1/2+x,1/2+y,z)	2.53362
Mn1-Al1(1/2+x,-1/2+y,z)	2.53362
Mn1-Al1(1/2+x,y,1/2+z)	2.53362
Mn1-Al1(-1/2+x,y,1/2+z)	2.53362
Mn1-Al1(1/2+x,y,-1/2+z)	2.53362
Cr1-Ni1(1/2+x,y,1/2+z)	2.53362
Cr1-Ni1(1/2+x,-1/2+y,z)	2.53362
Cr1-Ni1(-1/2+x,1/2+y,z)	2.53362
Cr1-Ni1(1/2+x,1/2+y,z)	2.53362

Distance

Bond Def.	Distance, Å
Cr1-Ni1(x,1/2+y,-1/2+z)	2.53362
Cr1-Ni1(x,-1/2+y,1/2+z)	2.53362
Cr1-Ni1(x,1/2+y,1/2+z)	2.53362
Cr1-Ni1(1/2+x,y,-1/2+z)	2.53362
Cr1-Ni1(-1/2+x,y,1/2+z)	2.53362
Cr1-Fe1(1/2+x,-1/2+y,z)	2.53362
Cr1-Fe1(1/2+x,y,-1/2+z)	2.53362
Cr1-Fe1(-1/2+x,y,1/2+z)	2.53362
Cr1-Fe1(1/2+x,y,1/2+z)	2.53362
Cr1-Fe1(-1/2+x,1/2+y,z)	2.53362
Cr1-Fe1(1/2+x,1/2+y,z)	2.53362
Cr1-Fe1(x,1/2+y,-1/2+z)	2.53362
Cr1-Fe1(x,-1/2+y,1/2+z)	2.53362
Cr1-Fe1(x,1/2+y,1/2+z)	2.53362
Cr1-Mn1(-1/2+x,y,1/2+z)	2.53362
Cr1-Mn1(1/2+x,y,-1/2+z)	2.53362
Cr1-Mn1(x,1/2+y,1/2+z)	2.53362
Cr1-Mn1(x,-1/2+y,1/2+z)	2.53362
Cr1-Mn1(x,1/2+y,-1/2+z)	2.53362
Cr1-Mn1(1/2+x,1/2+y,z)	2.53362
Cr1-Mn1(-1/2+x,1/2+y,z)	2.53362
Cr1-Mn1(1/2+x,-1/2+y,z)	2.53362
Cr1-Mn1(1/2+x,y,1/2+z)	2.53362
Cr1-Cr1(x,1/2+y,-1/2+z)	2.53362
Cr1-Cr1(1/2+x,y,-1/2+z)	2.53362
Cr1-Cr1(-1/2+x,y,1/2+z)	2.53362

Distance

Bond Def.	Distance, Å
Cr1-Cr1(1/2+x,y,1/2+z)	2.53362
Cr1-Cr1(1/2+x,-1/2+y,z)	2.53362
Cr1-Cr1(-1/2+x,1/2+y,z)	2.53362
Cr1-Cr1(1/2+x,1/2+y,z)	2.53362
Cr1-Cr1(x,-1/2+y,1/2+z)	2.53362
Cr1-Cr1(x,1/2+y,1/2+z)	2.53362
Cr1-Al1(x,-1/2+y,1/2+z)	2.53362
Cr1-Al1(1/2+x,y,-1/2+z)	2.53362
Cr1-Al1(-1/2+x,y,1/2+z)	2.53362
Cr1-Al1(x,1/2+y,1/2+z)	2.53362
Cr1-Al1(1/2+x,y,1/2+z)	2.53362
Cr1-Al1(1/2+x,-1/2+y,z)	2.53362
Cr1-Al1(-1/2+x,1/2+y,z)	2.53362
Cr1-Al1(1/2+x,1/2+y,z)	2.53362
Cr1-Al1(x,1/2+y,-1/2+z)	2.53362
Al1-Al1(x,1/2+y,1/2+z)	2.53362
Al1-Al1(x,-1/2+y,1/2+z)	2.53362
Al1-Al1(x,1/2+y,-1/2+z)	2.53362
Al1-Al1(1/2+x,1/2+y,z)	2.53362
Al1-Al1(-1/2+x,1/2+y,z)	2.53362
Al1-Al1(1/2+x,-1/2+y,z)	2.53362
Al1-Al1(1/2+x,y,1/2+z)	2.53362
Al1-Al1(-1/2+x,y,1/2+z)	2.53362
Al1-Al1(1/2+x,y,-1/2+z)	2.53362